Josquin Guerber

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Education : Ecology and Evolution

2018–2023 **École Normale Supérieure de Paris**, *Biology Department*

2022 **Master of Science in Biodiversity, Ecology and Evolution**, *University of Montpellier*

2019 **Bachelor in Life Sciences**, *École Normale Supérieure de Paris*

2016 **French-German Baccalaureate**, *Lycée Franco-Allemand de Buc*

Research Experience

2023–2026 **What are the links between community stability and long-term population trends ?** PhD student under the supervision of Élisa Thébault (iEES, SU, Paris) and Colin Fontaine (CESCO, MNHN, Paris). *Community ecology, citizen science data analysis, theoretical ecology.* Co-animated the "Ecology and Evolution" working group and co-animated skillshare sessions.

2023 (6 mos.) **Functional diversity of common bird communities.** Research internship, supervised by Mathilde Vimont, Charles Thévenin and Benoît Fontaine (CESCO, MNHN, Paris). *Community ecology, data analysis, biodiversity indicators*

2022 (5 mos.) **Stability of a trophic chain with overexploitation.** Research internship, supervised by Isabelle Gounand and Nicolas Loeuille (iEES, SU, Paris). *Modelling, theoretical ecology*

2022 (5 mos.) **Spatial structure of global drylands vegetation.** Research internship, supervised by Sonia Kéfi (ISEM, Montpellier). *Ecological stability, image analysis*

2020 (5 mos.) **Composition of benthic macrofaunal communities of the Barents Sea.** Research internship, supervised by Èric Jorda Molina and Henning Reiss (Nord University, Bodø, Norway). *Marine community ecology*

Positions Held

2023–2026 **PhD student,** Muséum National d'Histoire Naturelle, Paris.

2018–2023 **Student civil servant,** École Normale Supérieure - PSL, Paris.

Publications

preprint **Guerber, J.**, Genettais, D., Fontaine, C., and E. Thébault. *Population and community variability deviate from stationary expectations during transient dynamics.* bioRxiv. 10.64898/2026.07.08.737188

2026 **Guerber, J.**, Fontaine, C. and E. Thébault. *Population trends and variability within bird communities are amplified by intense land use.* **Journal of Animal Ecology**. 10.1111/1365-2656.70254

2026 **Guerber, J.**, Loeuille, N. and I. Gounand. *Overexploitation can counteract top-down control and the paradox of enrichment in simple food chains.* **Oikos**. 10.1002/oik.11470

2025 Pichon, B., Kéfi, S., Gounand, I., Gross, N., Le Bagousse-Pinguet, Y., **Guerber, J.**, Eldridge, D.,..., Donnet, S. and F.T Maestre. *Grazing Modulates the Multiscale Spatial Structure of Dryland Vegetation.* **Global Change Biology**. 10.1111/gcb.70345

2024 Kéfi, S., Génin, A., Garcia-Mayor, A., Guirado, E., Cabral, J.S., Berdugo, M., **Guerber, J.**, Solé, R. and F.T. Maestre. *Self-organization as a mechanism of resilience in dryland ecosystems.* **Proceedings of the National Academy of Sciences** (PNAS). 10.1073/pnas.2305153121

Teaching

2023–2026 Teaching assistant, MNHN (32h/year). *Microbiology practicals (Master's, 12h/year), scientific mediation in plant biology, botany and ecology (20h/year).*

2018–2019 Instructor in the TalENS program, ENS (30h total). *Scientific mediation for high schoolers.*

Student supervision

2026 (2 mos.) First year Master's internship of Zélie Andrieu Decoster, Sorbonne University. *Conservation ecology, data analysis.* Co-supervision with Laurence Walch (iEES, Paris).

2025 (5 mos.) Gap year Master's internship of Damien Genettais, ENS. *Theoretical ecology, stochastic differential equations.* Co-supervision with Colin Fontaine and Elisa Thébault.

2024 (2 mos.) First year Master's internship of Melaine Delattre, Sorbonne University. *Theoretical ecology, competitive Lotka-Volterra model.* Co-supervision with Elisa Thébault.

Presentations at international conferences

2025 How do population and community stability change during increasing or decreasing population trends ? **British Ecological Society Annual Meeting.** Edinburgh.

2024 Birds in farmland exhibit increased population variability beyond the impact of species population trends. **International congress of the French Society of Ecology and Evolution** (SFE2). Lyon.

Complementary training

Pedagogy **Teaching in higher education** (MESR, 2024), **Teaching and pedagogy for diverse audiences** (MNHN, 2024)

Methodology **Theory-driven analysis of ecological data** (CESAB, 2024)

Software tools

Programming	R, Julia, Python, bash
Software	git/GitHub, SLURM, Docker
Writing	Quarto, LaTeX

Languages

French	native
English	fluent
German	advanced